"What Does a Game-Trained Embedding Learn?" (draft of 8 Oct 2026)

Summary of the paper as it stands, and the review comments that matter

Prepared for P. Sole-Vilaro and coauthors, 26 Oct 2026. Every number comes from the reproduction bundle or from the 120-run baseline ensemble (R1) in `rev2/`; the full assessment and run plan are in `REVISION_PLAN.md`.

1. The paper in brief

The paper takes the multi-head architecture of Tarancón-Álvarez et al. (2026), which embeds a family of PDE solutions in a shared latent curve read by per-instance linear heads, and trains it by adversarial self-play on a pursuit–evasion game on the flat torus. It then asks what the latent space of the trained network encodes. The answer is organised around separating two objects: the *strategy ensemble* (the heading curves the heads output, one per evader initial condition), which game theory constrains, and the *latent curve* $\boldsymbol{H}(t)$ upstream of the heads, which it does not. The main claims are (i) the ensemble is low-rank, as the constant-heading saddle predicts, and this rank is selected by training rather than built in; (ii) the latent curve carries a larger "overhead" that no head reads and that is attributable to the input featurisation; (iii) the game's topology appears as clusters in control space, not as latent dimensions; (iv) a tempting pursuer–evader spectral asymmetry does not survive a symmetrised-training control. A self-play protocol study (§8) and a value-level check against an independently trained HJI critic (§9) complete the draft. Every claim has a named control with a pre-stated decision rule (Table 1 of the draft).

Setup

- **Game.** Pursuer P ($v_P = 1.2$) and evader E ($v_E = 1$) on $\mathbb{T}^2 = [0, 2)^2$; capture when the min-image distance $d_{\mathbb{T}} \le \varepsilon = 0.08$ within $t_{\max} = 4$. Feedback value $V^*(\boldsymbol{\Delta}) = (d_{\mathbb{T}}(\boldsymbol{\Delta}) - \varepsilon)_+/(v_P - v_E)$; saddle open-loop controls are constant headings along the shortest displacement. Forced-capture reach $R = (v_P - v_E)t_{\max} + \varepsilon = 0.88$.
- **Architecture.** Per player: Fourier features of time ($K = 12$ harmonics) → tanh body → latent $\boldsymbol{H}_\alpha(t) \in \mathbb{R}^{16}$ → $N_h = 16$ linear heads → unit headings $\hat{u}^j_\alpha(t) = W^j\boldsymbol{H}/\|W^j\boldsymbol{H}\|$. Open-loop: the policy depends on t only; the pursuer starts at the origin.
- **Training.** Differentiable soft capture time $\widehat{T}$, a one-sided no-capture penalty ($\lambda_{nc} = 2$) and distance shaping $\lambda_d\langle d_{\mathbb{T}}\rangle$ ($\lambda_d = 5$); pursuer warm-up, then 14,000 alternating iterations at 3:1 steps, Adam $10^{-2}/10^{-3}$. Nothing analytic enters any loss ("honesty contract"), with one disclosed exception: on $\mathbb{T}^2$, $d_{\mathbb{T}} = (v_P - v_E)V^* + \varepsilon$ off the capture set, so shaping is value shaping; it carries ≈ 100% of the evader's gradient and 6.4% of the pursuer's.

2. Claims, evidence and review verdict, section by section

Section	Claim in the draft	Evidence in the draft	Review verdict
§3 Dimensionalities	Strategy-ensemble PR(Z) = 1.5–3.0 "across every protocol variant", matching the saddle prediction ≈ 2; latent PR($\boldsymbol{H}_E$) = 5.4 ± 0.9; PR($\boldsymbol{H}$)/PR(Z) ≥ 1.69 in all 43 checkpoints; the two populations do not overlap.	One baseline run, 7 init seeds + 5 IC seeds (C1), 21 snapshots of 4 symmetrised runs.	**Needs restating.** PR(Z) is gauge-dependent (§3.1 below); the saddle bounds a different statistic (Gram PR ≤ 2, not "ideal 2"). The latent-vs-policy separation is real (≈ 2×, in 99–100% of 120 runs) but "≥ 1.69 in every checkpoint" fails at scale (minimum 1.49). "Populations do not overlap" holds only for the legacy pair.
§3 Stronger play	Oracle-trained pursuer has PR ≈ 2.4–2.7 on capturable heads; rank two is a property of constant-heading play, not of this weak equilibrium.	3 seeds; 4 capturable heads; head-centred PR.	**Underpowered** ($n = 3$, 4 curves). Rerun with 10 seeds and varying ICs (plan item R10).

Section	Claim in the draft	Evidence in the draft	Review verdict
§3 Warm start (C7)	A planted rank-9.3 heading family collapses to rank 2–3 within 8k iterations of game training.	One run under a wall-clock-chunked protocol.	**Confirmed on the checkpoint** with the corrected statistic (Gram PR $10.7 \to 2.3$, constancy $0.12 \to 0.89$), but $n = 1$; this is the claim that keeps the rank result from being trivial, so it needs 10 seeds (R4).
§4 Featurisation	Latent spectrum breaks at $k \approx K$ and tracks K (C3); overhead unrelated to winding-class count (C4); variance split 4.7:1 init : IC (C1).	C3: one run per K; C4: 8 runs; C1: $n = 8$ vs $n = 5$.	**Mixed.** Break at K confirmed at $n = 100$ (98% of latent power at $k \leq K$). The variance claim is *reversed*: at 10×10 seeds the init explains 0% and the IC set up to 65%; the remainder equals the spread from 10^{-10} perturbations. The featurisation attribution remains correlational; the decisive test is training without a game (R3).
§5 Topology	Winding classes (signed, $\mathbb{Z}^2$) are tight clusters in control space (6× separation), invisible to spectral statistics (Prop. 1); a reflection control shows the class is signed.	16 heads, 12 classes, 5 within-class pairs, one an outlier.	**Circular.** The winding label equals round$(2\bar{u})$ on 99% of heads, so it bins the quantity whose distances are measured; a permuted label gives 1.0×. The non-circular discrete content is *which image chart* each head plays (§4 below).
§6 Δv sweep	Neither dimensionality tracks the capturability break at $\rho = 1$.	8 values × 3 seeds, evader only, fixed ICs, legacy statistic.	**Qualitatively fine, under-sampled**; redo with both players, 10 runs per value, varying ICs (R8).
§7 Asymmetry	Raw head spectra are gauge artefacts (Lemma 2); a 2× pursuer excess of band power $k \in [11, 50]$ survives gauge fixing but becomes seed-dependent under symmetrised training, so no inf–sup attribution.	4 symmetrised seeds reported (8 exist in the bundle).	**Robust at population level under the baseline protocol** (ratio 1.86, ≥ 1.5 in 81% of 100 runs) *but carried by post-capture motion*; its mechanism, not its existence, is the open question (R2, R5). Report all 8 C2 runs.
§8 Protocols	The 23%-capture actor is limited by co-adaptation, not capacity: an oracle opponent converges, GDA/ABR cycle, OGDA/EG/Cesàro/EMA fail; Shapley locates the gap in the $k \in [1, 4]$ band.	4 protocols × 3 seeds; 6 corrections × 3 seeds; 3 Shapley seeds.	**A second paper**: about self-play optimisation, not the embedding. Move to an appendix; keep only the oracle arm. Its premise ("mediocre") is mis-measured (§3.5 below).
§9 Value check	Critic feedback policy captures 31% of the capturable test set vs 23% for the actor; actor–critic alignment factorises through the saddle direction; no latent-to-value bridge (R^2 0.46 vs 0.53 from time alone).	Critic from the companion paper (its smooth, Fourier-feature version).	**Replace.** V^* is closed form on $\mathbb{T}^2$; the critic adds nothing there, and the 23/31% measure generalisation to untrained ICs. On its own ICs the actor captures at $T_c/V^* = 1.01$. Keep the latent–value null.
§10–11	13 references; limitations and five future directions.	—	Related work must grow to ≈ 50 references; limitations should list the actual limits (one game, one geometry, open-loop, $N_h = 16$, $n = 3$–8).

3. The review's main findings (not in the earlier LLM evaluation)

3.1 The headline statistic is not gauge-invariant. $\hat{u}^j = z^j/\|z^j\|$ is unchanged by $z^j(t) \to c_j(t)\, z^j(t)$, $c_j > 0$; Lemma 2 states this for constant c_j. The paper's PR(Z), the time-centred PR of the pre-normalisation outputs z^j, changes under it: on the published baseline a smooth $c(t)$ moves PR(Z_P) from 1.915 to 1.121 while the policy, the temporal PR of $\hat{u}$ and the Gram PR stay fixed. Across 100 runs the pursuer's PR(Z) (median 1.75) sits well below its gauge-invariant counterpart (3.35), so the reported "pursuer ≈ 2" reflects head-output magnitudes. The replacements, each with a formula in `rev2/metrics.py`, are: the *Gram PR* of the heading curves as elements of L^2 (the object Proposition 1 bounds by 2); the *constancy* $\eta = \frac{1}{N_h}\sum_j \|\langle \hat{u}^j \rangle_t\|^2$; and the *temporal PR of* $\hat{u}$ for the time-varying part.

3.2 Training is chaotic, so seeds are not identifiers. Runs are bit-identical on one machine, but adding 10^{-12} to the initial weights gives $\max|\Delta B| = 0.81$ after training. The published seed-0 baseline gives PR($\boldsymbol{H}_E$) = 6.00 on this machine against 6.75 published. Consequently single-run numbers are draws from a distribution, and "variance is set by the initialisation" conflates initialisation with chaos. The honest reproducibility statement is: distributions reproduce; single runs reproduce only on identical hardware and software.

3.3 The control arms used incompatible protocols. Iteration counts are 10k (C1), 14k (baseline, C4, C5, Δv), 15k (C3) or 28k (C2); the Adam-reset cadence is 2,500 or 3,500 iterations, or a 33-second wall-clock chunk (C1 init seeds, C2, C7) that cannot be reproduced. The "eight init seeds" combine one 14k/3,500 run with seven 10k/33-s runs, and the IC arm is a third protocol. R1 shows the reset cadence itself barely matters, but the census must be made uniform.

3.4 The baseline IC set is unrepresentative. Capturable heads for IC seeds 0–19: 4, 7, 9, 9, 8, 12, 7, 9, 7, 10, 11, 8, 12, 7, 10, 8, 11, 5, 9, 6 (mean 8.45; expected 8.8). Seed 0 is the least capturable of the twenty, and every "capturable heads" statistic in the draft rests on four curves.

3.5 "Mediocre equilibrium" is mis-measured, and the real result is better. The 23% (and the critic's 31%) come from `capture_time_map`, which plays the nearest trained head's open-loop controls from each point of a 24×24 grid of untrained ICs. On its own ICs the self-play pursuer captures 92% of capturable heads at $T_c/V^* = 1.010\,[1.004, 1.015]$ (100% at the baseline), i.e. it realises the feedback saddle. At the same time a best-responding pursuer that knows the evader's committed path intercepts all 1,600 heads, including the 780 "uncapturable" ones, by $t \le 0.86$, mostly through a different image chart. Self-play finds a *chart-local* equilibrium of an open-loop game that has no value; it is not a Nash equilibrium. Exploitability is exact and closed-form on $\mathbb{T}^2$.

4. What the new baseline ensemble shows (R1: 120 runs, one deterministic protocol)

10 init seeds $\times$ 10 IC sets at the published configuration, plus a chaos arm (10^{-10} perturbations) and a no-reset arm. Median [IQR] over the 100-run grid; the published run's percentile in parentheses.

Statistic	Evader	Pursuer	Published run
Gram PR (saddle value for the same IC sets: 1.92)	2.47 [2.32, 2.70]	2.50 [2.30, 2.69]	2.54 / 2.48 (59th / 46th)
Constancy η, full horizon	0.852	0.848	0.85 / 0.87
Constancy η, *before* capture (captured heads)	0.9997	0.9998	
Constancy η, *after* capture	0.18	0.15	
Temporal PR of $\hat{u}$ (gauge-invariant)	2.80 [2.39, 3.24]	3.35 [2.85, 3.78]	2.85 / 3.14
Legacy PR(Z)	2.43 [1.95, 2.74]	1.75 [1.57, 2.04]	2.99 / 1.91 (91st / 65th)
PR($\boldsymbol{H}$), time-centred (legacy)	5.73 [5.29, 6.13]	6.51 [5.98, 7.28]	6.75 / 7.08 (**98th** / 67th)
PR($\boldsymbol{H}$) / temporal PR of $\hat{u}$, within run	2.04 [1.77, 2.30]	1.96 [1.71, 2.29]	
Band-power ratio P/E, $k \in [11, 50]$	1.86 [1.59, 2.16]; ≥ 1.5 in 81% of runs		1.92 (56th)
Capturable heads of 16	8.5 [7, 9]		4 (**5th**)
Capture rate on capturable heads; T_c/V^*	92%; 1.010 [1.004, 1.015]		100%
Best-response pursuer: heads captured	1,600 / 1,600 (all by $t \le 0.86$)		16 / 16

- **During play the strategies are constant saddle headings to four decimals.** All of the ensemble's excess over rank 2 is motion *after* capture, where only the shaping term still acts (close-range pursuit at 1.2 vs 1.0). The draft's "soft-game corrections" are post-game motion. The asymmetry statistic is likewise carried by captured heads (1.86 vs 1.50 on uncaptured heads).
- **The game's discrete content is the chart choice** (Fig. 2). Every head plays a pure saddle in some

image chart (cos with that chart's saddle direction: median 0.99999, minimum 0.978); both players choose the same chart on 99.7% of heads; the pursuer plays the nearest chart m^* on 92% of capturable heads. Capturable heads in m^* are captured 99.9% of the time, heads in another chart 3%: 62 of the 63 capture failures are chart errors, and the error rate falls from 48% at the cut locus to 1% far from it. This replaces §5 and ties directly to the chart construction of paper 2.

- **Variance.** Two-way ANOVA: the init seed has no detectable effect on any of 14 statistics ($p = 0.10$–0.82); the IC set is the systematic factor (Gram PR $p < 10^{-9}$; IC means track the number of capturable heads, Spearman 0.85); for the latent PR the residual is indistinguishable from the chaos floor. Resampling the draft's C1 design from the grid gives a median init:IC variance ratio of 0.73 and $P(\text{ratio} \geq 4.7) = 3.5\%$.
- **Protocol.** Adam resets change nothing in the ensemble or game statistics ($p \geq 0.27$, paired); population medians converge by 3,500 iterations, while individual runs keep wandering (run-level Spearman between 10.5k and 14k: 0.4–0.7), so distributions, not single runs, should be reported.

5. Where the review agrees with the earlier evaluation

Verified numerically: the capturable share is $\pi \cdot 0.88^2/4 = 60.8\%$, not 59.7%; the 4.7:1 variance ratio is not significant ($F(6, 4)$, $p = 0.078$); the winding label is a heading bin, not a homotopy class; Proposition 1 bounds the uncentred Gram PR, for which 2 is an upper bound, not an ideal; the run census (43 "runs") is 25 runs plus snapshots; C3 was trained to 15k, not 14k. Agreed on the text: §8 is a separate study; §9 should verify against closed-form V^* and reduce the critic to a pointer; "multi-head PINN" is wrong because no PDE residual enters the loss; the prose needs a human rewrite from figures and tables, an abstract under 250 words, and a related-work section of about 50 references (a vetted list is in `REVIEW_claude.md`, §7). Not checked here: the claims about paper 2's drafts and the citation corrections (Isaacs; Melikyan–Ovakimyan 1993; Comm. Phys. 2025). One disagreement: the "aggressive" timeline should not keep the 8-seed baseline, because those seeds mix protocols; reruns cost about an hour.

6. Recommended changes and the remaining runs

Where	Change
Abstract, §3	Lead with: before capture, game-trained strategies are the constant-heading saddle of a jointly chosen image chart ($\eta = 0.9997$, $\cos = 0.99999$). Drop PR(Z); report Gram PR, η and the temporal PR of $\hat{u}$ as distributions over the grid.
§4	Keep the overhead (latent temporal modes $\approx 2\times$ policy modes in 99–100% of runs) and the break at K; replace the init/IC split by the ANOVA; make the featurisation attribution interventional (R3).
§5	Replace the winding analysis by chart selection: misses are chart errors concentrated at the cut locus.
§7	The asymmetry is robust but post-capture; decide its mechanism (R2) before any inf–sup reading (R5).
§8, §9	§8 to an appendix (oracle arm stays). §9 becomes closed-form verification: $T_c/V^* = 1.01$; chart-local, fully exploitable equilibrium.
Throughout	One deterministic protocol; ≥ 10 runs per arm; a figure of the game, the architecture and the image charts; reproducibility stated as "distributions reproduce".

Remaining runs, in order of usefulness (about 10 laptop-hours in total; each has a decision rule in `REVISION_PLAN.md`): **R2** capture-truncated loss, paired with R1 (tests the post-capture mechanism behind the rank excess and the asymmetry); **R3** game-free training of the same architecture (tests "featurisation residue"); **R4** warm-start collapse at 10 seeds; **R5** symmetrised protocol and single-factor ablations for the asymmetry; **R6** K and body-width sweeps; **R7** heads sweep $N_h \in \{8, 32, 64, 128\}$; **R8** Δv sweep, both players, 10 runs per value; **R9** shaping sweep; **R10** oracle arm and the evader side of exploitability.

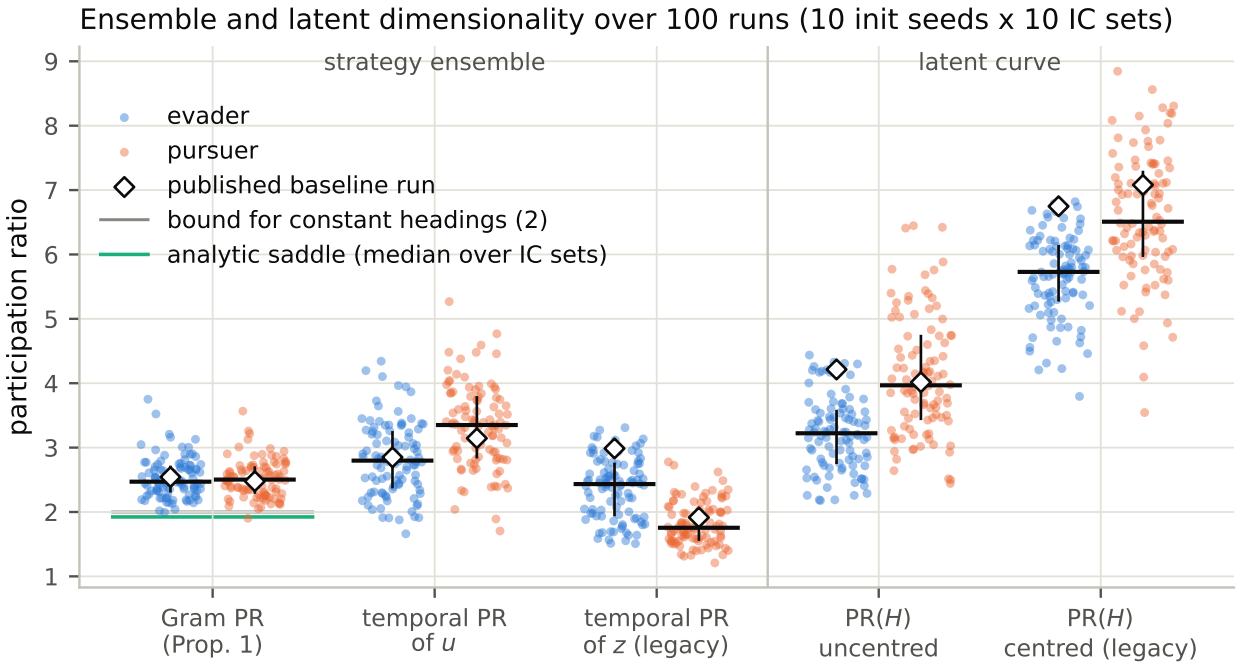


Figure 1: Ensemble and latent dimensionality over the 100-run grid, with the published run marked. Left of the divider: gauge-invariant Gram PR and temporal PR of $\hat{u}$, and the legacy PR(Z). Right: latent PR, centred and uncentred.

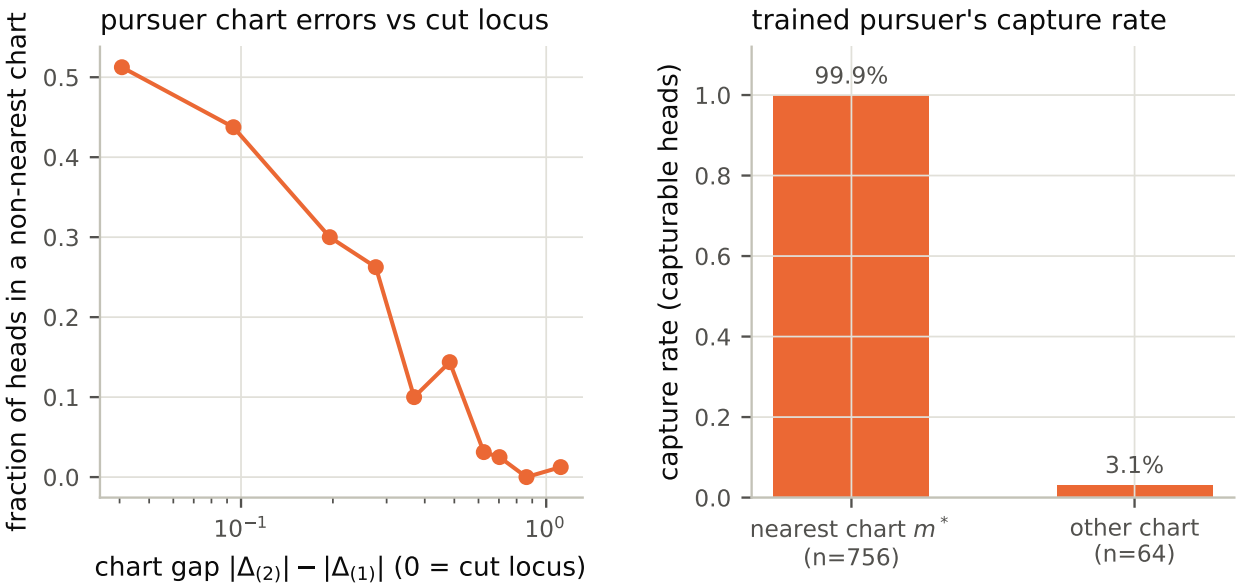


Figure 2: Chart selection. Left: fraction of heads playing a non-nearest image chart against the chart gap $|\mathbf{\Delta}_{(2)}| - |\mathbf{\Delta}_{(1)}|$ (deciles; the gap is 0 on the cut locus). Right: the trained pursuer's capture rate on capturable heads by chart played.